

Milestone 3 Report: Model Training, Evaluation, and Analysis

1. Introduction

This milestone focuses on the implementation, training, and evaluation of an object detection model using the YOLOv8 architecture. The objective was to analyze the performance of different model variants and understand their effectiveness on a medical imaging dataset. The work builds upon the preprocessing pipeline developed in earlier milestones and aims to generate meaningful insights from model predictions.

2. Model Selection and Training

Two variants of the YOLOv8 model were used:

- **YOLOv8n (Nano)** – Lightweight and faster model
- **YOLOv8s (Small)** – More complex with higher learning capacity

Training Configuration

The models were trained using the following parameters:

- Epochs: 60
- Image Size: 512 × 512
- Batch Size: 2
- Optimizer: Default (SGD/Adam based on YOLOv8)
- Device: NVIDIA RTX 3050 GPU (4GB VRAM)
- Workers: 0 (to ensure compatibility with Windows)
- Mixed Precision: Disabled
- Caching: Disabled

The dataset consisted of:

- Training Images: 4000
- Validation Images: 1000

3. Training Observations

During training, the following trends were observed:

- Gradual decrease in **box loss, classification loss, and DFL loss**
- Stable convergence without overfitting
- Consistent improvement in evaluation metrics across epochs

The training process for YOLOv8s took approximately **8.7 hours**, which is acceptable given hardware limitations.

4. Performance Evaluation

Final Metrics (YOLOv8s)

- Precision: ~0.48
- Recall: ~0.30
- mAP@50: ~0.336
- mAP@50-95: ~0.174

Comparison with YOLOv8n

Model	mAP@50	Training Time
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YOLOv8n	~0.34	~8 hours
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YOLOv8s	~0.336	~8.7 hours
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Analysis

Although YOLOv8s is a more complex model, it showed only marginal improvement over YOLOv8n. This indicates that increasing model complexity alone does not significantly enhance performance for this dataset.

5. Prediction and Visualization

The trained model was tested on validation images using the prediction mode. The model generated bounding boxes on detected objects and saved results in the output directory.

Observations from predictions:

- Some images showed **accurate detections**
- Several images had **no detections**
- Some predictions included **false positives**

6. Error Analysis

The prediction results revealed several limitations:

1. Missed Detections

- The model failed to detect objects in low-contrast images
- Indicates difficulty in learning subtle patterns

2. False Positives

- Incorrect detections in noisy or complex regions
- Suggests over-sensitivity to texture variations

3. Low Recall

- Many actual objects were not detected
- Indicates incomplete feature learning

4. Dataset Challenges

- Medical images have complex structures
- Variability in annotation quality may affect performance

7. Key Insights

- Model performance is **limited by dataset complexity**, not just model architecture
- Increasing model size does not guarantee better accuracy
- Proper preprocessing and high-quality annotations are crucial
- YOLOv8 is capable but requires domain-specific tuning for medical data

8. Conclusion

In this milestone, object detection models were successfully trained and evaluated using the YOLOv8 framework. While the models demonstrated stable learning and reasonable accuracy, performance improvements were limited due to the challenging nature of the dataset.

The results highlight the importance of dataset quality and domain-specific challenges in medical image analysis. Future improvements should focus on better preprocessing techniques, data augmentation, and annotation refinement rather than solely increasing model complexity.

9. Future Work

- Improve dataset quality and labeling accuracy
- Apply advanced data augmentation techniques
- Experiment with larger models (YOLOv8m/l) using cloud GPU
- Tune hyperparameters for better recall
- Integrate model into a real-time detection system

10. Summary

This milestone successfully demonstrates:

- End-to-end model training
- Performance evaluation
- Comparative analysis
- Practical understanding of deep learning limitations

The work lays a strong foundation for further improvements and real-world deployment.